EquiSplit — Video Demo Recording Script

EXACT 3-MINUTE WALKTHROUGH

Comprehensive spoken script & screen action guide for End-Term Project Submission

Presenter: Pranjay Srivastava | **Target Duration:** 2 mins 45 secs to 3 mins | **Project:** EquiSplit (MERN Expense Splitter)

1. Browser Setup
Have `http://localhost:5173` open in full screen (or 1080p). Logged out on the Auth page.

2. Audio & Pace
Speak with energetic confidence. Keep a steady pace. Pause 1 second before clicking buttons.

3. Key Objective
Highlight the *Min-Cash-Flow Greedy Algorithm* and *Penny-Accurate Remainder Math*.

Scene 1: Introduction & The Core Problem

0:00 – 0:30 (30S)

VISUAL ACTION:
Start on the **AUTH / LOGIN PAGE** showing the hero header and the demo account login pills.

SPEAK WORD-FOR-WORD:

"Hello everyone, my name is **Pranjay Srivastava**, and this is **EquiSplit** — an intelligent expense-sharing and debt-simplification platform built with the MERN stack."

"When roommates or travel groups share costs, typical apps create a tangled web of circular debts where everyone owes everyone, plus fractional penny rounding errors. EquiSplit solves this with mathematical precision and the **Min-Cash-Flow Greedy Algorithm**."

Rubric Covered: Problem Definition & Product Motivation

Timing Check: At 0:28, click "Rahul Sharma (Organizer)" to log in.

Scene 2: Dashboard & Group Architecture

0:30 – 1:00 (30S)

VISUAL ACTION:
Show Rahul's dashboard. Point cursor to aggregate balance cards (+₹3,150.00 you are owed). Click on **"GOA ROADTRIP 2026"** card.

SPEAK WORD-FOR-WORD:

"Logging in as Rahul, the dashboard immediately displays our net financial position across all groups: total spent, amount owed to us, and amount we owe."

"Opening our **Goa Roadtrip 2026** group, notice the clean header with our custom currency and a unique **6-character invite code**: `GOA202`. Membership is strictly protected by JWT authorization."

Rubric Covered: Architecture, Authorization & Natural State Display

Timing Check: Reach Group Detail by 0:58.

Scene 3: Algorithmic Highlight — Min-Cash-Flow Simplification

1:00 – 1:45 (45S)

VISUAL ACTION:

Click the **"BALANCES & SETTLE UP"** tab. Show the 2 simplified transfer cards. Click **"SHOW ALL RAW PAIRWISE DEBTS"** toggle to show the difference.

SPEAK WORD-FOR-WORD:

"Now let's examine the core algorithm: **Debt Simplification**. In a four-person trip, pairwise expenses generated **7 complex mutual debts**."

"EquiSplit's backend executes a greedy Min-Cash-Flow algorithm in $O(N \log N)$ time. By calculating net balances and iteratively pairing the maximum debtor with the maximum creditor, it collapses those 7 chaotic transactions down to just **2 optimal payments** — saving 71% of bank transfers!"

Rubric Covered: Engineering Rigor & Algorithm Implementation (Key 20% mark)

Timing Check: Switch back to Expenses tab at 1:43.

Scene 4: Precision Multi-Split Engine & Penny Remainder Logic

1:45 – 2:20 (35S)

VISUAL ACTION:

Click **"+ ADD EXPENSE"** button. Modal opens. Enter:
- Description: `Beachside Snacks`
- Amount: `100`

Show the Equal split distribution (₹33.34 + ₹33.33 + ₹33.33). Briefly click "Exact" and "Percentage" tabs to show live error validation. Click "Save Expense".

SPEAK WORD-FOR-WORD:

"Next is our precision **Multi-Split Engine**. Watch what happens when we split a ₹100 bill 3 ways. A naïve division gives non-terminating decimals."

"EquiSplit calculates shares in base paise and allocates remainder cents to ensure the sum **strictly equals 100.00** with zero rounding loss. We also support multi-payers, exact amount validation, and percentage splits that enforce an exact 100% total."

Rubric Covered: Mathematical Edge Cases, Form Validation & Reactive UI

Timing Check: Modal closes and new expense appears in list by 2:18.

Scene 5: Settle-Up, Confetti & Safety Guards

2:20 – 2:45 (25S)

VISUAL ACTION:

Switch to **BALANCES & SETTLE UP** tab. Click **"SETTLE"** on the card (Aman → Rahul). Settle modal opens. Click **"RECORD ₹1,000 PAYMENT"**.

Watch celebratory confetti trigger!

Switch to **MEMBERS TAB** & attempt to remove an indebted member to demonstrate the blocked action.

SPEAK WORD-FOR-WORD:

"With one click on **Settle**, Aman pays Rahul. The transaction instantly updates the ledger, recalculated balances, and triggers a celebratory animation."

"Furthermore, EquiSplit enforces **natural safety guards**: you cannot remove a member if they have an active non-zero balance — preserving ledger integrity at all times."

Rubric Covered: Business Guards, Settlement Logic & Delightful UX

Timing Check: Confetti plays at 2:32; safety guard shown at 2:42.

Scene 6: Analytics, Audit Trail & Conclusion

2:45 – 3:00 (15S)

VISUAL ACTION:

Click **"ANALYTICS"** tab (show category breakdown & progress bars), then click **"AUDIT TRAIL"** tab (show chronological timeline of all creations and settlements).

SPEAK WORD-FOR-WORD:

"Finally, our **Analytics** tab offers visual category spending breakdowns and Paid vs. Consumed ratios, while the immutable **Audit Trail** records every transaction."

"EquiSplit is backed by **21 passed automated tests**, MongoDB persistence, and a modern glassmorphic interface. Thank you!"

Rubric Covered: Audit Trail, Analytics & Professional Closing

Target End Time: 2:58 to 3:00 sharp!

Quick Reference Demo Accounts (One-Click Pills on Login Page):

User	Email	Password	Role in Demo Data
Rahul Sharma	`rahul@example.com`	`password123`	Organizer & Villa Payer (+₹3,150.00 credit)
Priya Patel	`priya@example.com`	`password123`	Roommate Admin (+₹2,100.00 credit)
Aman Verma	`aman@example.com`	`password123`	Trip Member (-₹1,000.00 debt to Rahul)